Introduction

DE-452: Cloud Computing
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Course Web Link: https://sites.google.com/pucit.edu.pk/muhammadabdullah/cloud-computing-spring-2023

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hadoop

amazon web services™

Spark

mongoDB

python™

scikit learn

Quizzes

- We will have few quizzes to check that you're keeping up with the material between exams
- Quizzes may be unannounced or announced in advance
- If you happen not to attend class on the day a quiz is given, you will get a 0 for that quiz

Honesty Policy

- Taking someone else's work and representing it as your own is lying, cheating, and stealing.
- Any reference material taken from the Web or other sources must be properly cited.
- Violations of the honesty policy will, at the very least, result in no credit for the work and 0 sessional marks.

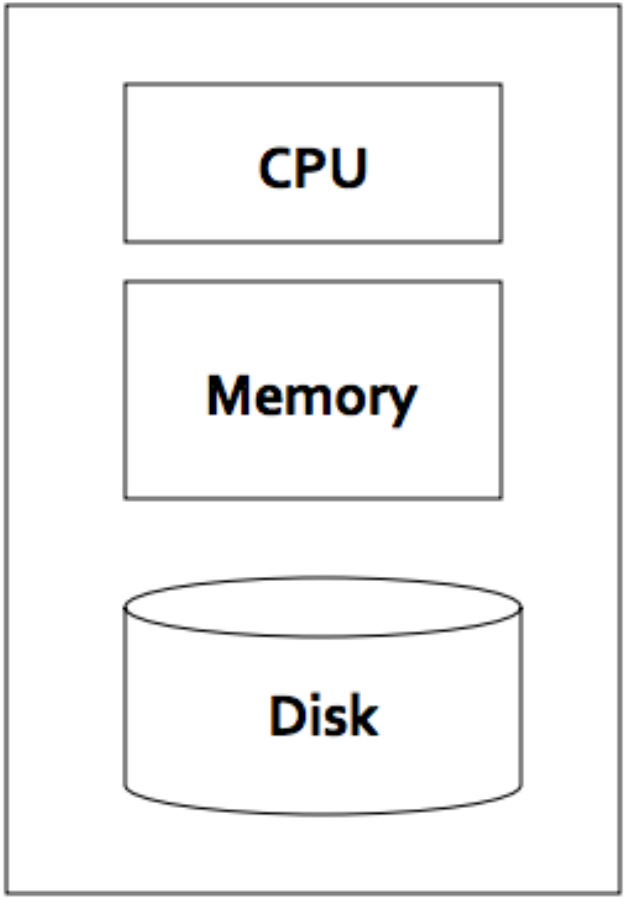
Cheating on Homework Assignment

- Copying from another or copying from the Internet are considered as cheating.
- BOTH PARTIES participated in the cheating will be penalized

Attendance and Late Arrival

- You are expected to attend every lecture and be on time.
- If you must come in late for unavoidable reasons, take extreme care not to disturb the class.
- Anyone arriving after I finish taking attendance will be counted as absent for that day.

Single Node Classic Architecture



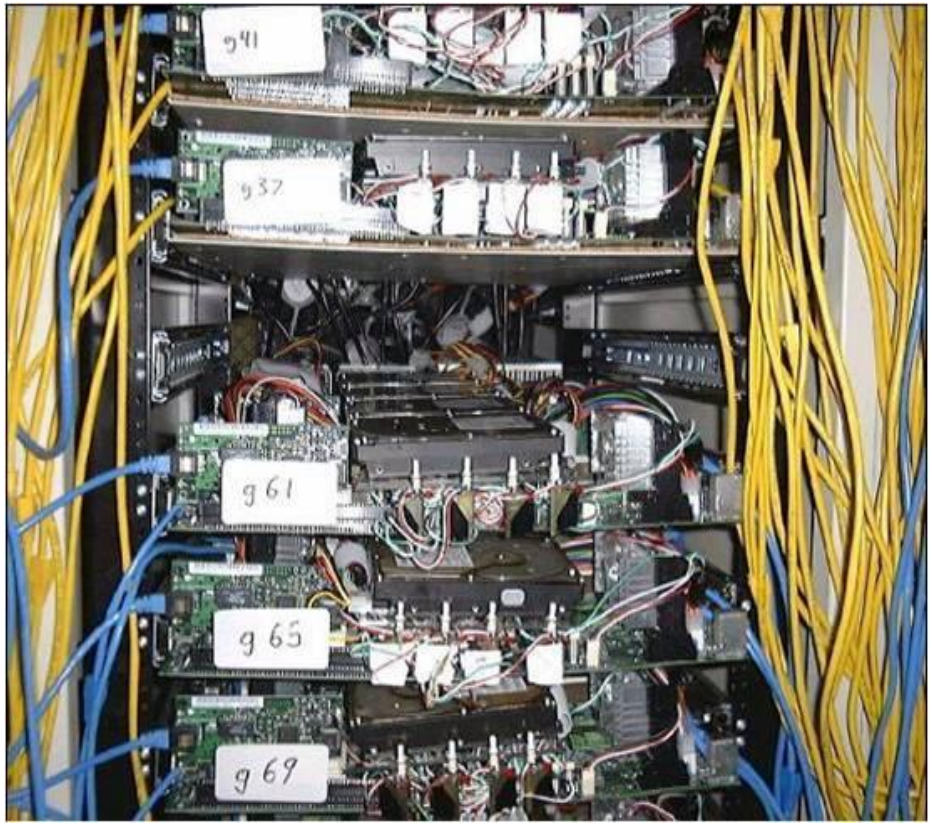
An Example of Google

- **10 billion** web pages
- Average size of webpage = **20KB**
- 10 billion * 20KB = **200TB**
- Disk read bandwidth = **50 MB/sec**
- Time to read = 4 million seconds = **46+ days**
- Even longer to do something useful with the data

Google 1998

Google was founded in 1998 by Larry Page and Sergey Brin while they were Ph.D. students at Stanford University in California.

Google 2001



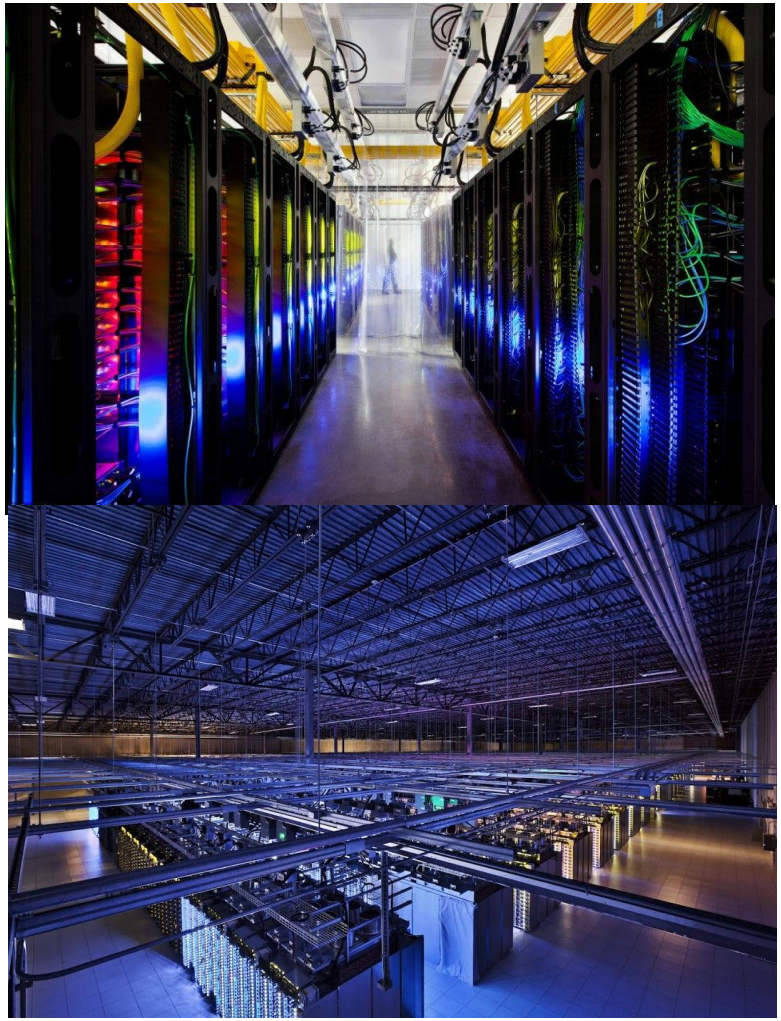
Commodity CPUs

Lots of disks

Low bandwidth network

Cheap !

Google Now!



The need of cloud computing?

- Individuals and small business cannot afford clusters
- Here comes the need of cloud computing!

Data Growth

- Google (as of around 2009) processes around 24 petabytes of data every day
- This is quite a lot, how much? Lets try to visualize the scale of data!

Let's imagine that a single byte is represented by a single grain of rice

1K or 1024 bytes would a bowl of rice

1 MB

1 GB 2 containers

1 TB 2048 Shipping Containers

Cloud Computing and Big Data

- In this course, we will be discussing tools, technologies, and concepts related to Cloud Computing and Big Data fields!

Home Task

- Install Ubuntu **18.04 LTS (Bionic Beaver)** on your machine!
- Build an image quality conversion script in any of your favorite programming language. The idea is to reduce size of the image without loosing significant quality of the image!
- **Hint:** ImageMagick can reduce the quality of a JPEG file with a simple command:

```
$ convert -quality 70 original.jpg optimized.jpg
```

- How would you sure that 70 is appropriate for a specific image!

Original Image
Size: 196KB

Optimized Image
Size: 100 KB

What is next!

- Introduction to Cloud Computing!